MIL-R-3911D
2 October 1985
SUPERSEDING
MIL-R-3911C
23 February 1978

MILITARY SPECIFICATION

RAIL, TEE, RAILWAY: RELAYER RAIL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers relayer rail suitable for use on railroads, marine railroads, and crane runways.

1.2 Classification. Relayer rail shall be classified according to its nominal weight.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of the standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2250

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN RAILWAY ENGINEERING ASSOCIATION (AREA)

AREA Manual for Railway Engineering, Vol. 1, Page 4-2-6.6.

(Application for copies should be addressed to the American Railway Engineering Association, 2000 L Street, NW, Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 1 - Carbon Steel Tee Rails.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovenment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Relayer rails shall be used rails suitable for use in trackwork. Unless otherwise specified, the rails shall have originally conformed to ASTM A 1. When rails originally conforming to specifications or standards other than ASTM A 1 are furnished, the contractor shall furnish the contracting officer certified proof that the rails originally conformed to the cited specification or standard. Relayer rails shall be free from split or crushed heads, piped or split webs, split or broken bases, or kinks or crooks which will prevent their being spiked to line and gage when laid. Unless otherwise specified, rails shall be of one section and weight per yard. Rails shall meet the requirements specified in table I.

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TABLE I. Relayer-rail characteristics.

Characteristics	Nominal weight less than 100 pounds	Nominal weight of 100 pounds and greater
Min. original weight per yard	Equal to the nominal weight	Equal to the nominal weight
Wear:		
Max. top wear 1/	see 3.8	See 3.8
Max. side wear 2/	see 3.8	See 3.8
Length	80% of lot 27 ft (8 meter) and longer; 20% of lot between 20 ft (6 meter) and 27 ft (8 meter) long.	90% of lot 30 ft (9 meter) and longer; 10% of lot between 22 ft (7 meter) and 30 ft (9 meter) and long.
Defects permitted:		
Headburn	Max. 1-1/2 in. (38 mm) dia.; Max. 1/16 in. (2 mm) deep; Max. 6 per rail.	Max. 1-1/2 in. (38 mm) dia.; Max 1/16 in. (2 mm) deep; Max 6 per rail.
Max. lip 3/	3/32 in. (2 mm)	1/8 in. (2 mm)

1/ Top wear shall be measured at the center of the rail.
2/ Side wear shall be measured at the side of the head, 5/8 in. (16 mm) below the original top of the rail and shall be the total on both sides.
3/ Lip is defined as flow of metal from the crown and shall be measured at the side or end or both of the rail.

3.2 Cropping. Rails with end defects shall be cropped by sawing off the defect. The rails shall not be cropped by the use of chisel or flame. After cropping, the length of the rail shall be in whole feet, ±3/8 inch (10 mm).

3.3 Drilling. All rails of the same section and weight per yard shall be drilled in accordance with ASTM A 1 unless more than one drilling pattern per rail section is authorized by the ordering data. A hole shall not be closer than 1-1/2 inches (38 mm) to the edge of an old hole.

3.4 Base width. The base of rails of the same section and weight per yard shall be of uniform width.

3.5 Length. Rails shall not be joined by welding in order to meet length requirements.

3.6 Identification marking. The letters "US", weight, section, and length of rail shall be stenciled with white paint on top of the rail approximately 3 feet (914 mm) from the end of each rail.

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3.7 Workmanship. Rails shall be free from burrs, fins, and sharp edges.

3.8 Rail wear. Unless otherwise specified, maximum top and side rail wear for the rail weight required shall conform to AREA Manual for Railway Engineering, Vol. 1, Page 4-2-6.6, Recommended Rail Grading Classification, Class II, Maximum Rail Wear - Inches. For rail weights required which are not listed on page 4-2-6.6, use maximum top and side wear for next smaller rail weight which is listed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Examination. Each rail shall be examined for defects specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. Each rail shall be examined for the following defects:

101. Length of rails not as specified (see 3.1, table I).
102. Rail section and weight not as specified (see 3.1, table I).
103. Rail boltholes not as specified (see 3.1, table I).
104. Rail wear beyond specified limitation (see 3.1, table I).
105. Rail section(s) not as specified (see 3.1, table I).
106. Headburns more numerous and larger then specified (see 3.1, table I).
107. Lip is larger than specified (see 3.1, table I).
108. Rail ends not cropped as specified (see 3.2).
109. Drilling pattern(s) not as specified (see 3.3).
110. Marking incorrect or illegible (see 3.6).
111. Workmanship not as specified (see 3.7).
112. Rail wear grading classification not as specified (see 3.8).
113. Base width not as specified (see 3.4).
114. Length requirements not as specified (see 3.5).

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4.5 Inspection of packaging.

4.5.1 Examination. The marking shall be examined for conformance to 5.1.

5. PACKAGING

(Rails shall be prepared for shipment as loose items.)

5.1 Marking. Loose rails shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

a. Rail of 100 pounds and greater is intended for use on main lines, access track, or other tracks where train movement may be classified as heavy or the desired speed is in excess of 40 mph (kmph).
b. Rail of less than 100 pounds is intended for use on yard, industrial storage, siding, and running tracks.
c. Rails intended for marine railways and crane runways should be specified according to the load requirements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Weight of rail required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Specifications or standards other than ASTM A 1 to which rails must conform (see 3.1).
e. Number of different rail sections permitted (see 3.1).
f. Number of drilling patterns permitted (see 3.3).
g. If not as specified, rail grading classification for rail wear (see 3.8).

Custodians:
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Navy - YD

Review activity:
Army - CE

Preparing activity:
Army - ME

Project 2250-0063

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-3911D	Rail, Tee, Railway: Relayer Rail

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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